

Zac Freeman
me@zac-freeman.com

Experience

Systems Engineer | AutoZone

January 2020 - Present

- Automated a manual provisioning process for OAuth clients with a GitOps system that managed the entire request lifecycle and provided a secure, auditable platform for AI-generated changes
- Led the transformation of a shared Gitlab CI/CD pipeline project from a single-purpose Java deployment tool to the enterprise-standard product for securely deploying any application to any environment
- Built a comprehensive automated test suite for our Gitlab CI/CD pipelines with a focus on easy extensibility to accelerate development without sacrificing reliability as the product surface scaled to ~50 pipeline types, multiple active versions, and over 2000 end-user Gitlab projects
- Rescued a project to replace the employee self-service site for 140,000 employees, releasing a working product on-schedule
- Updated a custom Qt/C++ time clocking software to prompt employees with surveys to meet California payroll compliance, recovering \$2,000,000 a year in penalties
- Performed the duties of contributor, reviewer, and trainer for C++ payment processing software handling \$5B a year across 20,000 devices

Software Developer II | Frontdoor

October 2018 - December 2019

- Led the software development for a security initiative, implementing reCAPTCHA for login and registration, session token handling, email verification, and ADFS SSO into a legacy ATG B2B site
- Integrated a 3rd party payment processor to meet new PCI standards for a legacy ATG B2B site
- Designed and built a microservice with Go and Docker to enable B2B clients to place orders in database without interacting with legacy ATG B2B site

Research Assistant | UAH Heliophysics

May 2017 - August 2017

- Created a model of gamma rays in a magnetic trap with the goal of reproducing real world data
- Used C++, CUDA, and the Runge-Kutta method to simulate many parallel gamma ray systems on various NVIDIA GPUs
- Analyzed data produced to discover novel behavior of cosmic rays inside expanding magnetic traps

Research Assistant | FSU Math Department

June 2016 - August 2016

- Used MATLAB and numerical methods for partial differential equations to simulate the flow of two liquids and analyze the physical accuracy of the data using energy conservation and the maximality principle
- Results from simulations contributed to the paper *Maximally Preserving Finite Difference Schemes for the Allen-Cahn Equation using Operator Splitting*, presented at SIAM-SEAS 2017

Education

B.S. in Physics | Florida State University

August 2014 - December 2017

- GPA: 3.889
- Minor: Mathematics

Skills

Languages	C++, Python, L ^A T _E X, Bash, Javascript, Redstone
Software	Linux, Qt, git, Docker, Gitlab CI/CD, Google ADK